



PCBoard®

Internet Collection Manual

Version 1.1

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1

Introduction

What Is The PCBoard Internet Collection?

The Internet has broadened our horizons. Not only has it helped us connect with other sites around the world in a matter of a few seconds, but it has also shown us how that can be done at a low cost.

The bulk of Internet services are provided by Internet Service Providers (ISPs). While they can connect you to the Internet, they cannot provide the sense of community provided on a PCBoard Bulletin Board System.

The Best Of Both Worlds

It makes perfect sense to combine the global outreach of the Internet with the sense of community and personal service provided by a BBS. That is why the PCBoard Internet Collection (PCB/IC) was created. With this software package popular Internet features such as World Wide Web, FTP, and Telnet are all available directly from your system.

Furthermore, your system still provides all of the features that made it popular in the first place: conferences, doors, local message bases. That is how both worlds can be merged to both your benefit and that of your callers.

Requirements

The PCBoard Internet Collection has requirements that must be adhered to in order to operate properly. The following details the requirements:

Dedicated Internet Connection

The Internet Collection requires you to have a permanent or dedicated connection to the Internet. There are a number of ways to get this type of connection such as T1, fractional T1, ISDN, and dial-up. These terms may not mean anything to you on first inspection, but your local Internet provider can explain the cost of each option and which is best for your situation.

Fractional T1 may be a good starting point if you do not already have a connection. The reason this is a good investment is that the hardware is basically the same for a full-T1 as a fractional. Therefore, as your system and needs grow, another investment is not required in hardware. However, smaller systems may

want to start with a dial-up connection. Call the providers in your area for pricing and technical details.

Once the Internet connection is installed, the signal must be put onto your network. Typically this is done in combination with a router. For specific requirements, contact your Internet provider.

TCP/IP

If you have any experience with the Internet, chances are that you have heard this term before. The term is an acronym that stands for **T**ransmission **C**ontrol **P**rotocol/**I**nternet **P**rotocol or TCP/IP. This term refers to the "language of the Internet" or the way that machines communicate with one another.

When TCP/IP is enabled on your network, a protocol or TCP/IP stack is maintained that allows the network and software run on the network to communicate with your hardware. In this case, the software is the PCBoard Internet Collection. More information on TCP/IP and how to add it to your system is discussed in the *First Time Setup* chapter.

The way that the TCP/IP stack is accessed varies between operating systems. PCB/IC works with DOS and OS/2 based TCP/IP stacks. From a DOS level, the PACKET DRIVER over **ethernet** networks method is used. For OS/2, the API is accessed directly.

PCBoard

The PCBoard Internet Collection is compatible with v14.5 or greater of PCBoard. However, several Internet related features have been added to more recent PCBoard upgrades. This is something to keep in mind as you evaluate the requirements.

Memory

PCB/IC only requires about 400k of memory to operate properly. Since this memory requirement is less than PCBoard requires, there will be no difficulty making it work.

Goals Of The Manual

Understanding the design goals of this manual and how the information is organized is designed to help you get the most out of this manual. When you know where and how to find information, this manual becomes a very valuable resource.

First, we make sure you understand the requirements of PCB/IC so that you know what you are getting into. This is an essential part of any undertaking because it helps you manage the expectations of the project. We've already gone over the requirements which means by now you can meet them and are prepared to move on.

Second, we talk about how to install the software for the first time. This is more than just copying files, but also how to customize the software to meet your needs and also how to secure the features so only desired users may access them.

In the next few chapters we cover topics that relate to the configuration and maintenance of PCB/IC. Topics covered include customization of the look and feel of the program, security and access considerations, etc.

The last part of the manual is dedicated mostly to reference material including descriptions of the programs in the collection and appendix chapters to cover any additional pertinent material.

2

First Time Setup

Establish A Feed

Some work must be done before you can put PCB/IC on-line. The two fundamental things that must be taken care of is to get a compatible TCP/IP stack installed and make sure it provides a permanent IP address.

TCP/IP Stacks

An effort to stress *compatible* TCP/IP stacks has been mentioned and this may make you wonder if your stack is compatible. TCP/IP is TCP/IP, right? Yes, it is, but how you access that protocol varies from operating system to operating system. For example, a different API is used to access TCP/IP under Windows NT and under OS/2. Currently, any DOS packet-driver or OS/2 based TCP/IP stack works fine with PCB/IC.

Permanent IP Addresses

Some connections to the Internet, such as PPP and SLIP, do not always have a guaranteed IP address. Instead, you get whatever IP address the provider wants to assign to your machine. This is a problem in relation to PCB/IC because anytime you provide Internet access to other callers, a permanent IP address is required.

Put The Feed On The Network

Just because you have a feed, it does not mean that you are ready to go, yet. Steps must be taken to get that feed to be broadcast into your network. Typically this involves broadcasting multiple packet types. This means your network needs to be capable of supporting multiple protocols.

For DOS users, your ethernet network must support the PACKET DRIVER specification. If your network supports either NDIS or ODI (which most do), simply install the appropriate packet driver. This process is sometimes referred to as a shim.

Most OS/2 users will be using the TCP/IP product directly from IBM or utilize the built-in TCP/IP (Warp Connect / Warp Server). Some work may be required in making that product work with your existing network. Once that part is setup, however, the Collection talks directly to the TCP/IP API.

The next sections talk about some possible solutions with the most popular DOS based networks: LANtastic, NetWare, and Windows For Workgroups.

LANtastic for DOS

Artisoft provides NDIS drivers for their products. Once these NDIS drivers are installed, you can load NDIS packet-driver to complete the TCP/IP connection.

NDIS drivers ship with their current product. If you do not have them, they can be obtained from Artisoft's BBS (520-884-8648) and FTP site (<ftp.artisoft.com>)

Netware

Using the ODI implementation of NetWare, you can use an ODIPKT driver that is freely available on many Internet sites. You may also need additional packet drivers for your card. The *Crynwr packet driver collection* provides a good place to find packet drivers for most cards. This collection is obtainable from most FTP sites under the filename DRIVERS.ZIP.

OS/2

The OS/2 version of PCB/IC talks directly to the OS/2 TCP/IP API. There is nothing special you have to do other than to verify that your network under OS/2 does have TCP/IP enabled.

Windows For Workgroups

PCB/IC does not directly interact with the Windows For Workgroup's interface. As a result, Windows NT is not a viable platform for using PCB/IC. However, Windows 3.1 and Windows 95 can be used because you can still load DOS TCP/IP drivers in your CONFIG.SYS and AUTOEXEC.BAT files with another network operating system (e.g., LANtastic, NetWare, etc.) If you choose this method, you cannot use the Windows 95 network drivers too.

The IP Environment Variable

PCB/IC uses a configuration file called TCPIP.DAT that keeps track of many things such as IP addresses, security levels and so forth. The only node dependent field in the entire file is the IP address.

Because so much of the data is common between nodes, you can share one TCPIP.DAT for all machines by setting an environment variable named **IP** to have the IP address for the current node. For example, to set the IP address to be 198.42.22.16, use the following line:

```
SET IP=198.42.22.16
```

This setting needs to be in the batch file that runs each node (e.g., BOARD.BAT or NODEx.BAT) as it is used to differentiate which nodes get which incoming packets. If all nodes use the same IP address, PCB/IC will not operate properly.

In the case where one batch file is used to launch all of your nodes, use several statements to set the variables. Each statement checks a preset node environment variable to set the IP address. Example:

```
IF %PCBNODE%==1 SET IP=198.42.22.16
IF %PCBNODE%==2 SET IP=198.42.22.17
IF %PCBNODE%==3 SET IP=198.42.22.18
IF %PCBNODE%==4 SET IP=198.42.22.19
```

By setting the IP address for each machine via environment variables, you can share one TCPIP.DAT. If you choose not to use one data file, you must maintain a separate copy of TCPIP.DAT for each machine on your network.

With the data file location set in place, you are ready to begin the configuration process. First you must set the IP addresses for your machine and for other machines you access.

A Note About IP Addresses

In order to use an IP address, you have to be authoritative for it. That means that an Internet provider somewhere along the line has assigned you a set range of IP addresses to use. Using addresses outside of that range will result in the program not being able to successfully send or receive information around the Internet.

One other area to be concerned about when assigning IP addresses to your nodes is to make sure that the following are true:

1. The IP address is assigned a host name in the domain name server (DNS) registered to your domain. For example, people could contact the computer with this IP address by the name **computer1.site.com**. The name of the computer gets resolved to the actual IP address.

2. A lot of sites do what is called a *reverse DNS lookup* when your users contact them. If they cannot resolve your IP address into a name such as `computer3.site.com`, they will reject the connection. Make sure that the DNS entries for your IP addresses are in the reverse name look-up table. Contact the party responsible for your DNS if you are unsure.

The Ethernet Card Address

OS/2 users need to set one additional environment variable in order to use PPP and SLIP. This variable, named `ETHADDR`, defines the physical address of the ethernet card in the machine.

In your `CONFIG.SYS`, add a line which reads:

```
SET ETHADDR=nn:nn:nn:nn:nn:nn
```

In this example `nn` is a two digit pair of numbers from the 12-digit address. The following steps show you how to get the ethernet address configured:

1. Run `NETSTAT -n`. Write down the 12 digit physical address listed (e.g., `00006e24beb0`).
2. Break the number into two digit pairs. This address would be listed as `00 00 6e 24 be b0`.
3. Now just put the results into the `ETHADDR` variable by adding a line to `CONFIG.SYS` that reads `SET ETHADDR=00:00:6e:24:be:b0`.

Remember, the ethernet address varies from machine to machine, so the `netstat` program must be used to guarantee accuracy.

Configuring IP Addresses

Sites on the Internet are referenced by name or by what is called an *IP address*. The address is a series of 4 numbers separated by periods such as `205.128.64.32`. There are a few addresses you need to set up that allows PCB/IC to operate properly.

These addresses are configured by loading `PCBICCFG.EXE`. Once `PCBICCFG` is loaded, select *IP Addresses*. Here you will find 7 IP addresses that must be entered in order to use all of PCB/IC's features.

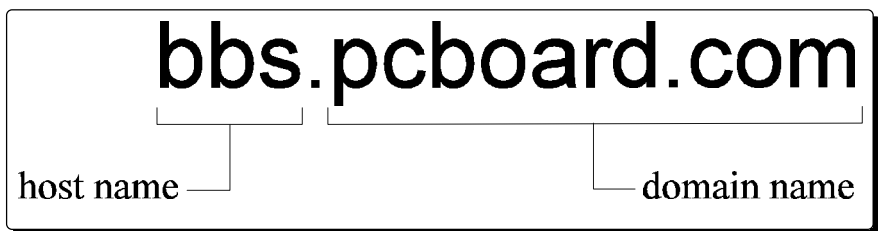
IP Address: First is the IP address for the current machine. This address is set via the IP environment variable as outlined in the previous section. To change this value, the node batch file must be changed as appropriate.

Net Mask: This is not an address but rather a mask to the address. The number you enter here is used in conjunction with any other IP addresses that are used to communicate with other sites on the network. The mask is used to determine if the machine is part of your subnet or not. If you have only one IP address (such as a SLIP or P3 connection) the mask typically used is 255.255.255.255. Class C sites that have access to the last 255 numbers use a mask of 255.255.255.0. The best person to talk to about your mask is your Internet provider as they know the exact mask to use.

Gateway: With the mask, we determine if a packet or datagram is destined for our subnet or group of computers. Packets that are not for us need to be sent somewhere. Where do these packets get sent? The answer to that question is an address referred to as the *gateway*. Typically, this is a router on your network that forwards packets based on its routing table.

Name Servers: There is room to enter IP addresses for 3 name server (DNS) machines. Most likely your provider has their own name server which they are more than happy to provide the address of. Additional servers can be obtained with the information obtained by the WHOIS application.

Domain Name: This is an optional field that you can fill in to contain the domain name for your system. For example, if I have registered site.com, I enter that value in this field. The domain typically ends with .COM, .GOV, or .EDU.



A sample internet address with domain name and host name clearly labeled.

Host Name: The name you enter in this field is typically the name for the machine. Many sites or domains have numerous hosts. That is, they have a www host, ftp host, etc. Realize that this name in conjunction with the domain

name makes up your entire Internet name. Also keep in mind that this is an optional entry.

Nic Server: With the vastness of the Internet, someone has to keep track of all the addresses and who is assigned to what. This information is only used in the WHOIS application of the collection. A handful of sites around the world track this information. The default IP address for this field is 198.41.0.5 which is the address for InterNIC.

Gopher Server: This field has a default gopher server to use. You may change it to any server you desire.

Test The Connection

With the IP addresses all entered, you are ready to begin testing the connection. First of all, make sure you have exited PCBICCFG and saved changes on the way out. The next step is to use the TESTIC.EXE application (TESTIC2.EXE for OS/2 users). This program tests multiple aspects of your Internet connection including pinging another site.

You are first asked to Enter Site Name. What you enter here must be a valid **named** address like `sitename.com`, not an actual IP address. Once this information is entered, the test proceeds.

If the tests are successful, a message stating further instructions is printed at the bottom of the screen. In the event that the tests were not successful, you need to narrow down the problem.

Troubleshooting

There are 4 major areas to look at when troubleshooting why the test program failed: 1) is the proper packet driver installed?, 2) your IP address, 3) the gateway address, and 4) name server addresses.

Packet Drivers

If the message `NO PACKET DRIVER FOUND` is displayed, you do not have a compatible DOS packet driver installed. Focus your efforts on the network drivers. Remember, you need to have NDIS or compatible installed and then run a packet-driver on top of that.

Your IP Address

If your IP address is not correct for the configuration, you may receive a **Could Not Resolve** message. An incorrect IP address would be something that is not within your assigned IP range. For example, if your provider has given you IP addresses 198.100.10.1 through 198.100.10.10, and you enter 198.100.15.10 for your IP address, the test will fail.

The Gateway Address

The gateway is what allows packets that are not for your subnet to get off your network and to be routed to their proper location. When an invalid or incorrect number is entered for the gateway address, the packets cannot get off of your network and you will receive a **Could Not Resolve** message.

Name Servers

The first test pings an actual IP address while the second test tries to ping the name you enter. The purpose of the second test is to see if your name server is able to resolve a name into an IP address. If not, the message **Could Not Resolve** is printed. Check to make sure your gateway address is correct and that your name server addresses are valid as well.

Adding PCB/IC To PCBoard

The PCBoard Internet Collection is accessed as a door application from the PCBoard system. It also requires your user file to be updated with a new allocation unit to store information about which applications a user can access, etc. In this section, we guide you through updating the user records and provide some ways you can add PCB/IC as a door application to your system.

User Records

In modifying the user records, a new allocation must be added to the user file.

Updating User File

NOTE: This step requires all nodes to be down. If anyone has the USER file open, this step cannot be accomplished.

In the directory where you installed PCB/IC is a batch file named TPA.BAT. This batch file runs PCBSM.EXE and adds the new allocation to the users file. Because this is a global change to the USER file, all nodes must be down.

Change to your \PCB directory and run the TPA.BAT file. Example:

```
C:\PCB> C:\PCBIC\TPA.BAT
```

A screen appears that details what addition is being made to the user file. Simply press PageDown to begin adding the new data. When the process is complete, you are returned back to the command prompt. A sharing violation error message while trying to update the user file means that some other node or process has the USERS file open. Rectify the situation and run the TPA.BAT file again.

Adding PCB/IC As a Door

At this point, you are ready to add the collection to your system. You can use any of the three methods outlined in this section. The first method details how to set up the PCB/IC applications under one door. The second method talks about setting up each application as a separate door. Finally, the last method talks about making SLIP and PPP connections easier for your callers. Based on your needs, one or more methods may need to be used.

As One Door Application

The collection can be added just like any other door program in PCBoard. Recall that doors are defined in the conference configuration screen from PCBSetup. The following are sample door entries for DOS and OS/2 respectively (Note that the only difference is in the **Os2** column):

	Filename	Pwrd	Sec	Login	USER SYS	DOOR SYS	Shell	Os2	Charges Minute	Per Use	Path to DOOR Files
1)	PCBIC		0	N	Y	N	N	N	0	0	C:\PCBIC

	Filename	Pwrd	Sec	Login	USER SYS	DOOR SYS	Shell	Os2	Charges Minute	Per Use	Path to DOOR Files
1)	PCBIC		0	N	Y	N	N	Y	0	0	C:\PCBIC

Two items about this definition are worth mentioning. First, notice that the USERS.SYS column is marked. This is how PCB/IC gets the extended user information it needs to operate. Second, notice the security is set to 0. Extensive security and access control is built into PCB/IC. For more information on these features, see the security chapter of this manual.

The way we have shown the configuration in this section, takes advantage of the sample door batch files included when you installed PCB/IC.

The PCBoard manual contains many other samples and details about configuring doors on the system. Refer to that manual if you have questions about how to add a door application to the system.

Separate Door Applications

Another way that users can access the Internet applications is one at a time. For example, by using the FTP command on your main menu, they are taken directly to the desired application.

The setup for this type of scenario is identical to the setup for one door entry with the following exceptions:

- One door is required for each application you make available.
- In the door batch file, you must load PCBIC.EXE with the appropriate command line parameter. This parameter specifies the application to load.

We shift our focus to the line from the batch file that loads PCB/IC:

```
C:\PCBIC\PCBIC%2%.EXE
```

To load the FTP application, and only the FTP application, simply use the command line parameter as shown:

```
C:\PCBIC\PCBIC%PCBOS%2%.EXE /FTP
```

The entries for the doors themselves are the same only there are more of them. Look at this example:

	Filename	Pwrd	Sec	Login	USER SYS	DOOR SYS	Shell	Os2	Charges Minute	Per Use	Path to DOOR Files
1)	FTP		0	N	Y	N	N	N	0	0	C:\PCBIC
2)	TELNET		0	N	Y	N	N	N	0	0	C:\PCBIC

The only difference between the two batch files is that the first one uses the /FTP command line parameter and the second one uses the /TELNET parameter. A comprehensive list of command line parameters for PCB/IC follows:

- /FTP, /FINGER, /GOPHER, /PING, /RLOGIN, /SLIP, /PPP, /TELNET, /TRACE, /WHOIS



When your system is setup in this manner, users can specify the site to go to right on the command line (e.g., `ftp ftp.pcbboard.com`).

SLIP/PPP (World Wide Web Access)

Many users of the Internet have become accustomed to the World Wide Web interface. As a result, your users may want to utilize this same interface and can with PCB/IC and the SLIP or PPP application. The target of this section is to set up the system for users to get SLIP/PPP access faster in the logon process thereby simplifying their logon scripts.

Adding The SLIP and PPP Doors

Rather than make the user load the PCB/IC door and then select either PPP or SLIP access from the main menu, capitalize on the capability to load an application from a single door by passing command line parameters. Sample DOORS.LST entries may resemble the following:

	Filename	Pwrd	Sec	Login	USER SYS	DOOR SYS	Shell	Os2	Charges Minute	Per Use	Path to DOOR Files
1)	PCBIC		200	N	V	N	N	N	0	0	C:\PCBIC\
2)	PPP		200	N	V	N	ZZZ	N	0	0	C:\PCBIC\
3)	SLIP		200	N	V	N	ZZZ	N	0	0	C:\PCBIC\

The PPP and SLIP door batch files are already created for you and exist in the directory PCB/IC was installed into. If you are using the OS/2 version of PCBoard, make sure you enable the OS/2 flag in the DOORS.LST.

Adding The SLIP and PPP Languages

To facilitate a faster logon, we need to setup separate languages for SLIP and PPP access. This way, users see the following type of mini-menu when they logon:

- 1 - Logon To PCBoard
- 2 - SLIP Internet Connection
- 3 - PPP Internet Connection

By utilizing the multiple language feature of PCBoard, we can let the user select what they want to use at logon. During the logon process, a PPE checks to see if the user selected either the SLIP or PPP language. If so, the user is launched directly into the SLIP or PPP door.

Adding the language into your PCBoard setup requires the following steps:

1. Using PCBS^{et}up, make sure languages are enabled. This is found in *PCBS^{et}up | Configuration Options | System Control* with the question **Multi-Lingual Operation**. This option must be turned on.
2. Add descriptions for the new menus by going to *PCBS^{et}up | File Locations | Configuration Files*. Edit the *multi-language data file*.

	Language	Extension	Country	CodePage	Yes	No
1)	1 - Logon To PCBoard				Y	N
2)	2 - PPP (Internet)	PPP			Y	N
3)	3 - SLIP (Internet)	SLP			Y	N

3. Create the new PCBTEXT.PPP and PCBTEXT.SLP files. First check in *PCBS^{et}up | File Locations | System Files* to make sure you know what subdirectories PCBTEXT files are stored. Once you know this location, exit to DOS, change to that subdirectory and then run MKPCBTXT. When prompted for the filename to open, enter PCBTEXT.PPP and create the file when prompted. Repeat this step for the PCBTEXT.SLP file too.

Installing The PPE

The final step in simplifying the logon process for SLIP and PPP callers is to install the PPE that detects if the user selected one of the languages we just setup. The name of the PPE is RUNINET.PPE and it is located in the \PCBIC\SCRIPTS subdirectory.

Keeping in mind that we want the user to get to the PPP or SLIP door as soon as possible, the PPE is best run at the login script questionnaire. If we do it any earlier, the user will not have logged into the system and consequently we could not guard access to the door. Let's get it installed.

Load PCBS^{et}up and go into *File Locations | Questionnaires*. Your screen will have these fields at the top:

```
Name/Loc of Logon  Script Quest. :
Name/Loc of Logon  Script Answers:
```

In the first line, enter the full path and location of where the RUNINET.PPE can be found. Example:

```
Name/Loc of Logon  Script Quest. : C:\PCBIC\SCRIPTS\RUNINET.PPE
Name/Loc of Logon  Script Answers:
```

If you already have a PPE or display file in that field, you can still run the PPE here. The modifications that are required depends on whether you are running a display file or a PPE in this field.

Display File

In this case, simply edit your display file so that at the first column of the first line is an exclamation point followed by the full path and filename of the PPE to run. Example:

```
!C:\PCBIC\SCRIPTS\RUNINET.PPE

Welcome to the BBS, @FIRST@!

Here at ACME BBS, we have a lot of things to offer our callers
and look forward to serving your needs. Please let us know
if there is anything we can do for you.
```

PPE

When a PPE is running, we simply need to take advantage of PCBoard's ability to run a PPE in a display file. Set the logon questionnaire to display a file like this example shows:

```
Name/Loc of Logon  Script Quest. : C:\PCB\LOGON.RUN
Name/Loc of Logon  Script Answers:
```

Edit the questionnaire and put both PPEs on the first two lines of the file remembering to precede each filename with the exclamation point:

```
!C:\PCB\PPE\EXISTING.PPE
!C:\PCBIC\SCRIPTS\RUNINET.PPE
```



NOTE: If you need \$\$LOGON.BAT/CMD processing, you need to pass the /LOGON parameter to this PPE on the command line. Example:
C:\PCBIC\RUNINET.PPE /LOGON

Sample Script Files For Your Users

Included in the \PCBIC\SCRIPTS subdirectory are several sample scripts that you can modify and hand out to your customers. Also included are instructions for your users on how to connect up with your systems.

Because these are all sample files for Salt Air -- CDC's support system, you will need to go through the files and make sure that any references to our system are replaced with your system information. This includes items such as phone numbers, language numbers, and BBS names.

We have made an effort to support the three main operating systems for the IBM-PC platform -- Windows 3.1, Windows 95, and OS/2. The following section describes what files correspond to which operating systems:

Operating System	Sample Script	User Instructions
Windows 3.1	WIN31PPP.CMD	WIN31PPP.TXT
Windows 95	WIN95PPP.SCP	WIN95PPP.TXT
OS/2	OS2SLIP.CMD	OS2SLIP.TXT

The Maintenance Event

Maintenance? What kind of maintenance do you need with an Internet door? There are two areas that need to be addressed: 1) Deleting files from the user holding directories, and 2) Updating the top 15 sites (hotpick lists).

Holding Directories

Maintaining the user holding directories is only an issue if you let your users put files downloaded via FTP into a holding directory. If so, a maintenance event needs to run so that it can enforce the number of days files should be held in the holding directory (see *PCBICCFG* | *Profile Information* | *Max Hold Days*). **PCBICEVT** is the program that does all of the work in the event.

HotpicksMaintenance

Activities involved with maintaining the hotpicks database include creating the top fifteen lists, trimming the size of the list. Optionally, all sites can be automatically approved for usage if you choose not to screen sites. All hotpicks maintenance is done via the PCBICEVT program.

When using the PCBICEVT program, it needs to be able to find the TCPIP.DAT configuration file in order to operate properly. Therefore, when you run it in your event, keep in mind that it searches for the file in the following manner:

1. If the /FILE:[filename] parameter is passed on the command line, the filename specified is used as the TCPIP.DAT file. This type of access may be useful if you are running this program on a file server that does not have a local copy of PCB/IC running.
Example: PCBICEVT /FILE:F:\PCBIC\TCPIP.DAT /RANK:5
2. If the TCPIP environment variable exists, the file specified by this environment variable is used. For example, if this variable contains the value P:\PCBIC\TCPIP.DAT, that is the file that is used.
3. PCBICEVT searches the current directory and then every directory in your PATH= statement.

Maintaining The Hotpicks Database Size

Two command line parameters to the event program control what the size of the list will be. The first parameter `/PURGE:[days]` determines after how many days of not being visited must elapse before the site is removed from the database. For example, if you use `PCBICEVT /PURGE:30`, any sites that have not been accessed in 30 days are removed from the hotpicks database.

Considering that the database holds a maximum of 2500 entries, make sure that there is sufficient room to add any new sites. The `/KEEP` parameter determines how many sites to keep in the database. Specifying `PCBICEVT /KEEP:1000` keeps the top 1000 entries in the site. Since this parameter only keeps the top number of sites you specify, it must reorder the database before purging any sites. Therefore, */KEEP must be accompanied* by the `/RANK` parameter. See Creating The Top Fifteen Lists for more information on the `/RANK` parameter.

The two parameters—`/KEEP` and `/PURGE`—can be used together. Older sites are purged according to the number of days specified and then if the database still exceeds the number of records specified, the database is packed down to the appropriate size. Example:

```
PCBICEVT /PURGE:30 /KEEP:1000 /RANK:5
```

Creating The Top Fifteen Lists

The top sites for your system is updated when you run the `PCBICEVT` program with the `/RANK:[scale]` parameter. Rather than sort on the number of accesses the site has had, a sliding scale is used. This prevents a situation where a site is red-hot in popularity for a while and then dies off. How does this sliding scale work?

What you have to consider in the sliding scale is the number of sites in your database and the number specified after the `/RANK` parameter. For example, if you have 600 entries in the hotpicks database and use the `/RANK:5` parameter, popularity for your database would be as follows:

#1 Site	3000 popularity points (600*5)
#2 Site	2995 popularity points (599*5)
#3 Site	2990 popularity points (598*5)
#600 Site	5 popularity points (1*5)

Notice that the gap between the #1 and #2 sites is whatever you specify with the `/RANK` parameter. The larger the value you specify (1-20 is valid) the harder it will be for a site to become the #1 site. Conversely, the lower the number, the

more volatile your top-15 list will be. Typically a value between 5 and 10 is a good starting point and then you can adjust that value to get the results you desire (i.e., increase the value to stabilize the list or decrease the value to make it change more often). Example:

```
PCBICEVT /RANK:5 /KEEP:600 /PURGE:30
```

NOTE: The top-fifteen lists are automatically created (without sorting them), whenever you specify any of the hotpicks maintenance command line parameters.

Automatically Approving Sites

By default, PCB/IC requires new sites to be approved before they are considered for the top fifteen list. To bypass site screening, add the /FORCE parameter to the PCBICEVT program that is running in your event. Example:

```
PCBICEVT /RANK:5 /KEEP:600 /PURGE:30 /FORCE
```

Configuring The Maintenance Event

The event itself is very easy to configure. Simply add a new event in the EVENT.DAT file (*PCBSetup | Event Information*) which looks something like the following:

Node:	1	Event Editor							
	Act	OS/2	Mod	Batch File	Begin Time	End Time	SMTWTFSS	Date	Last Date
1)	V	N	S	ICMAINT	02:00	03:00	YYYYYYYY		00-00-00
2)	N	N	N				NNNNNNNN		00-00-00
3)	N	N	N				NNNNNNNN		00-00-00
4)	N	N	N				NNNNNNNN		00-00-00
5)	N	N	N				NNNNNNNN		00-00-00
6)	N	N	N				NNNNNNNN		00-00-00
7)	N	N	N				NNNNNNNN		00-00-00
8)	N	N	N				NNNNNNNN		00-00-00
9)	N	N	N				NNNNNNNN		00-00-00
10)	N	N	N				NNNNNNNN		00-00-00
11)	N	N	N				NNNNNNNN		00-00-00
12)	N	N	N				NNNNNNNN		00-00-00
13)	N	N	N				NNNNNNNN		00-00-00
14)	N	N	N				NNNNNNNN		00-00-00
15)	N	N	N				NNNNNNNN		00-00-00

Modes: E = Expedited S = Sliding F = Fido M = Mail Hour
 ESC=Exit AltI=Insert AltD=Delete AltR=Repeat AltS=25/50 F2=Edit AltU=View
 08:15:41 02-07-96 F1=help caps: OFF num: ON ins: OFF

This figure shows a simple event definition for the PCB/IC maintenance event.

As for the batch file, that too is quite simple:

```
@echo off
c:
cd\pcbic
pcbicevt /rank:5
```

Nothing requires that you have this utility run in a separate event. If you already have a maintenance event for your system that does activities such as packing the message base or users file, add the PCBICEVT program to that event. The only requirement for the program to run is to have access to your TCPIP.DAT file (which is why we change directories in our sample event batch file.)

3

Restricting Access / Security

The Internet Collection offers an abundant set of security options that give you control over who gets access to various components. Available security options can best be categorized in the following areas: 1) normal PCBoard security, 2) application security, and 3) group templates. We will spend the next few sections discussing each:

Normal PCBoard Security

Because the Internet Collection functions as a door from PCBoard, you get all of the security benefits of running a door. This means you can control access by password or security level directly from the DOORS.LST file in the configuration screen for any given conference.

With these global security options, access can be quickly enabled or disabled for all users on your system. For more specific control of who has access, you'll need to explore the security options in PCB/IC. The next two sections are dedicated to discussions of this area.

Application Security Via Menu Options

The Internet Collection is broken down into several different applications. Access to each application is controlled by security level in *PCBICCFG* | *Alternate Menu Configuration*.



Program	Sec Level
TraceRoute	00000000
FTP	00000000
Gopher	00000000
finger	00000000
ping	00000000
ppp	00000000
rlogin	00000000
SLIP	00000000
Telnet	00000000
Whois	00000000

This type of security presents the easiest way to guard a particular application from unauthorized access. Simply set the minimum security level required to access a particular application.

Obviously, there are other issues to contend with. What if I want to allow users to try out an application before they decide to buy access to my system? This type of security will not work in this situation. That is correct and that is where

you need to look at what is known as group security. Utilizing the next level of security, you can put demo users and subscribers in separate groups and give them different access accordingly.

Application Security Via Groups

What Are Groups And Profiles?

Groups are, as the name implies, a way to group people together and give them access based on which ones they are a member of. For example, you may set up a group named DEMO. Everyone that is a member of this group has access associated with a demo user. The DEMO group most likely has less time per application and may not have access to all applications.

The term profile refers to the definition of a group. What applications they have access to, how long they can spend in the applications, etc. We will spend more time explaining what settings are in a profile later in this document. For now, it is simply important that you understand what is meant by group and profile.

Configuring Groups And Profiles

Groups are setup by selecting the "Profile Information" option from the PCBICCFG Main Menu. When you select this menu option you see a screen like the following:



The initial screen when you edit profiles for the first time.

This screen shows us a composite view of what the user SYSOP has access to. Notice that, by default, this user has no access. That is because we need to grant

the access we want to give the user. This is done by giving the user the access to certain groups.

Groups? I don't have any groups. With a new configuration, you're not expected to. However, that can be quickly remedied by pressing ALT-A to add a group to the current user followed by ALT-A again to add a group to the "Available" list.

Adding New Groups

Adding a new group is really quite easy. Why? Because we describe the users of your system verbally.

Naming Your Groups

Simply come up with a name to define a group of users. Some examples may be:

- ▶ PAYING MEMBER
- ▶ PUBLIC USER
- ▶ VISITING SYSOP
- ▶ SYSOP

Notice that the names are all based on nouns. You can use them in sentences like the following:

- ▶ He is a PAYING MEMBER.
- ▶ She is a VISITING SYSOP.
- ▶ He is a SYSOP.
- ▶ He is a PUBLIC USER.
- ▶ She is both a VISITING SYSOP and a PAYING MEMBER.

As you give the user classification of your system some thought, you'll be able to come up with group names you want to use. Once we have the groups defined, we can begin assigning them to users. First things first, however.

Setting The Group Properties

Remember, we had pressed ALT-A twice from the User Profile screen to get to the point where you had to enter a name. Now is the time you enter the names

you want to define. Let's go through an example by defining the group named "SYSOP". The first screen you see is as follows:



Program	Time Limits	
TELNET	No Access	
RLOGIN	No Access	
FTP	No Access	
SLIP	No Access	Allow Holding : N
PPP	No Access	Allow Attach's : N
GOPHER	No Access	Hotpicks : N
FINGER	No Access	
WHOIS	No Access	Max Hold Days : 0
PING	No Access	Holding Limit : 0
TRACERT	No Access	

Setting Application Time Limits

The first thing to do is to determine how much time a user should have in each application (0 = No Access). Use the TAB key to move between the fields and enter the maximum amount of time you want the user to stay in each application on a DAILY basis. To limit the user to the normal PCBoard time limit, simply enter a -1 which gets translated to 1440 minutes (24 hours).

Since we are creating a group for the System Operator(s), set all of the time limits to -1 (1440) and enable all of the Y/N options.

Setting Group Options

On the right side of the screen is a few options relating to hotpicks and holding directories for FTP files. Once again, the TAB key will move between the fields and you can set the appropriate setting.

Allow Holding: When this option is enabled, the user can store files retrieved from an FTP site in a temporary holding directory. Remember, the global "Allow Holding" option must be enabled in *PCBICCFG | System Configuration*. Held files are stored in the location specified in the same PCBICCFG screen.

Allow Attaches: Users can choose to store retrieved files as file attachments in the message base when this option is enabled. There is no global option to be concerned about in regards to attachments as there is with the holding directory.

Hotpicks: With this option enabled, the list of top 15 sites for a particular application is available at the site name prompt by pressing T followed by ENTER. Hotpicks databases are maintained for the FTP, TELNET, RLOGIN, and GOPHER applications.

Max Holding Days: Files that are still in the user's holding directory are deleted by the event utility, PCBICEVT.

Holding Limit: Anyone that is offering their system as a storage bin needs some sort of limit. This is the case with PCBIC as you can limit the user to a certain size (specified in kilobytes) for their holding directory.

Saving The Group

Saving the group is really easy. Press ESC to exit and confirm that you want to save the changes.

Creating Additional Groups

Not a lot of effort is required to get additional groups created. It is more of a repetition process. If you look at your user profile screen, you see something like this:



After adding all of the groups (ALT-A from this screen), your screen should be more like the following:



Adding Groups To A Security Profile

Rather than have to manually add groups to every user on your system, you can apply groups to all users with a given security level. This is done by going to the User Profile screen from *PCBICCFG | Profile Information*. Press ALT-S to edit security profiles. The default profile to enter is the security level of the current user.

In our example setup, the SYSOP has a security level of 250. Once we are done setting up the profile for security level 250, every user with that security level automatically becomes members of groups in the security profile. This makes system maintenance easier because now you can change the access of entire security levels very quickly.

Let's create two security profiles. One for level 250 (SysOp) and one for VISITING SYSOPs who have become PAID SUBSCRIBERS.

Example: Creating A Profile For Security 250 (SYSOP)

From the User Profile screen, press ALT-S to select the security level and enter 250 (or whatever your SysOp security is). Notice that you are now at the Security Profile screen. Here you can add groups that everyone with a security level of 250 will have. Let's add just the SYSOP group by doing the following:

1. Press ALT-A to add a group to the security profile.
2. Cursor down to SYSOP in the "Defined Groups" column.
3. Press ENTER to add the group.
4. Press ESC to exit and return to the security profile screen.
5. Notice the time limits now reflect that of the SYSOP group.
6. Press ESC one more time to return to the user profile screen.

PCBoard Internet Collection --- Configuration			
User Profile Maintenance			
NAME: SVSOP	SEC: 250	DIR: (DISABLED)	REC: 1 of 2
Selected Groups		Program	Time Limits
DEFAULT		TELNET	1440
SVSOP		RLOGIN	1440
Record: 2 of 2		FTP	1440
		SLIP	1440
		PPP	1440
		Gopher	1440
		FINGER	1440
		WHOIS	1440
		PING	1440
		TRACERT	1440
		Allow Holding :	Y
		Allow Attach's :	Y
		Hotpicks :	Y
		Max Hold Days :	0
		Holding Limit :	0

The SysOp has unlimited access because he is a member of the SysOp group we defined.

The DEFAULT group is automatically applied to everyone. The SYSOP group, in this example, comes from the security level profile. You can add additional groups for just this user by using ALT-A from this screen. Any groups added in this manner have no effect on those with a similar security level.

Example: Creating A Visiting Sysop Member Profile (Sec. 90)

Our example for security level 90 puts a bit of a twist into the picture. Now we have a group of users who are both Visiting SysOps and Paid Members. How do we accommodate them? The answer is easy. Simply assign the groups to that security level and let PCB/IC handle the rest.

1. Go to the User Profile Screen (*PCBICCFG | Profile Maintenance*)
2. Press ALT-S to edit security profiles.
3. Enter 90 for the security level to edit.
4. Add (ALT-A) VISITING SYSOP and PAYING MEMBER to the security profile.
5. Exit and save.

It's that easy. We've now just assigned security level 90 the privileges of a VISITING SYSOP who is a PAYING MEMBER. This brings up an interesting point. With more than one group, how does PCB/IC resolve the differences between the two. This is called a composite profile

How To Understand Group Composite Profile

The rules for resolving the access when two or more groups are involved is really quite simple—the user gets the most access granted to them.

If they are a member of a group that gives 10 minutes of access to SLIP, and also a member of a group that gives 45 minutes, the user gets 45.

Summary

There is a lot of territory that we have not covered in this documentation regarding groups and profiles. Yes, we have covered the essentials, but there is a lot you can explore in the way of adding and deleting groups. Don't be shy when it comes to satisfying your curiosity.

4

Configuration

PCBICCFG.EXE is the program that is used to set up and customize the Internet Collection. Obviously, you should become familiar with this program and how you can use it to customize the program to meet your needs. Although you have already used PCBICCFG to setup PCB/IC, this chapter's focus is to cover each option in greater depth.

File Directories

There are only a small number of directories that you need to configure in *PCBICCFG | System Configuration*. This section talks about each of those sub-directories, and what files are stored in each one.

Name/Loc of ICPROFS.DAT

This data file contains information about the group profiles you define. It is typically small and should be shared among all nodes. Therefore, the best place to put this is on a shared-drive out on the network. That way, all nodes can access the same data file. The following is an example:

```
Name/Loc of ICPROFS.DAT : G:\PCBIC\DATA\ICPROFS.DAT
```

Name/Loc of ICLEVELS.DAT

As you know by now, you can simplify your configuration by making everyone at a particular security level a member of several groups. That way, you do not have to manually update hundreds, or thousands of users. Information about what groups are assigned to what security levels are stored in this file. It is best to store this file on the server so it is accessible to all nodes.

```
Name/Loc of ICLEVELS.DAT : G:\PCBIC\DATA\ICLEVELS.DAT
```

Location of Data Files

The subdirectory you specify in this field determines where the menu data file and external text files are stored. Remember, only the subdirectory is required. File information should be ignored. The following shows an example:

```
Location of Data Files : C:\PCBIC\DATA\
```

Location of Holding Dir

This subdirectory serves as a starting subdirectory for holding directories. All user holding directories are stored underneath the directory specified in this field. For example, entering C:\PCBIC\HOLD\ for the holding directory will yield subdirectories named something like C:\PCBIC\HOLD\0A3FQG.IC\ . The 0A3FQG.IC subdirectory is unique to one individual on the system. A user's holding directory can be determined by looking at the user's profile in PCBICCFG. An example entry for this field follows:

Location of Holding Dir : C:\PCBIC\HOLD\

Location of External Log

Logs are kept to track activity. PCB/IC has the ability to either store the information directly in your normal PCBoard caller logs, or you can send them to an external log file. External log files are written to the directory specified by this field.

Each node maintains a separate log file. The filenames for all of the logs begin with TCP and are followed by the node number. For example, node 21's external file is named TCP21. The following shows an example entry:

Location of External Log : C:\PCBIC\LOG\

Application Options

In *PCBICCFG | System Configuration*, there are a few fields that are specific to just a few applications. This section discusses those options and their effect on the applications of the PCBoard Internet Collection.

Email Site Name

Some of the applications in the collection such as FTP need an e-mail address for anonymous logins. This field allows you to specify the suffix for the e-mail address. The first part of the address, user name, is automatically generated by the program.

Whatever you enter for the suffix, it should usually begin with the mandatory @ sign used in all Internet e-mail addresses followed by the site the user's e-mail

address is sent to. Most often, this is the domain name for your BBS. An example entry for this field follows:

Email Site Name : @sitename.com

With this field entry and a user named JOHN SMITH logged in, the user's full Internet e-mail address is `john.smith@sitename.com`.

FTP

Allow Holding of Files

Holding a file provides a more permanent way for users to store files they have retrieved via FTP in case they do not have enough time to get them from the BBS on the original call.

This option must be enabled if you want to allow any users to store files in a temporary holding directory. A setting similar to this one is also in each group profile so that you can disable holding capabilities for certain groups of users. For example, if you want SysOps, and only SysOps, to have access to hold files, you have to enable this setting and also the setting in the SYSOP profile (if that is what you named it). If either setting is set to "N", the user will not be able to store files in the holding directory.

Unlike file attachments, this option puts the file into a holding directory on disk. Since the holding directory is separate from the message system, there is no overhead with one message per file, etc. Files are packed out of the holding directories during a system event using the PCBICEVT program.

Attach Conference Number

When allowing these files to be stored as file attachments, it is strongly recommended that you setup a separate conference for these file attachments to be stored. That way, PCBPack can be used to pack-out these files in a different manner than other message bases.

Suppose you specified this conference number to be a message base that holds regular messages too (e.g., Main Board message base). In this scenario, there are two problems with storing the file attachments here:

1. The message base could become excessively large. This would be the case if users download a lot of files since each message attaches only a single file.

2. In normal conferences, you may want to leave messages online for 30-60 days before packing it off the system. If you let 30-60 days of file attachments pile up, a good chunk of disk-space is used in the process.

The answer to each of these problems is to set up a separate conference that has one purpose—store file attachments. In an event, set PCBPack to run with the /DAYS switch to remove attachments that are a certain age.

SLIP

SLIP IP Display Type

With the Internet Collection, you can change the way the initial screen displayed to SLIP callers. You may need to change this setting to make it easier for your user to create scripts. For example, some SLIP Internet applications display the host address first, while others display the client IP first. The following lists each of the options available to you and shows a sample of each:

Display Type	Example Output
1	Your IP address is: xxx.xxx.xxx.xxx
2	Annex (Gateway) address is: xxx.xxx.xxx.xxx
3	Your IP address is: xxx.xxx.xxx.xxx Annex (Gateway) address is: xxx.xxx.xxx.xxx
4	Annex (Gateway) address is: xxx.xxx.xxx.xxx Your IP address is: xxx.xxx.xxx.xxx
5	Use a custom display file (see "Creating a Custom Display File") in this section.

Creating a Custom Display File

If your needs dictate a format that is not provided, create a file named **IPDSP** in the node subdirectory or the data file subdirectory (*PCBICCFG | System Configuration*). Be careful if you choose to put this file in the data subdirectory because it contains node specific information (IP addresses) and is therefore not sharable among all nodes.

With this display file, all IP address information must be put in there by you. The Internet Collection simply displays the file to the caller. This may require some sort of external program to determine the caller's IP address and the gateway address.

Logging Options

Logging information proves useful in many circumstances. By viewing activity logs, you can tell whether a user accessed a particular application and more. PCB/IC has numerous options that control where log files are kept and what information is stored. These settings are found in *PCBICCFG | System Configuration*.

Use External File

Logging can be done in one of two ways—internal or external. Internal logging is where all entries are placed directly in the PCBoard caller log while external logging is where all PCB/IC log entries are kept in a separate file for each node. This option only affects the location of the entries themselves, not how much is tracked. Therefore, your selection here is simply based on whether you mind having larger PCBoard caller log files or prefer to have PCB/IC log entries in an external file.

When external logging is selected, the location for the log files is defined in *PCBICCFG | System Configuration*. The filenames all begin with TCP followed by the node number. For example, if this is your entry for the log locations,

```
Location of External Log : C:\PCBIC\LOGS\
```

the log for node 22 is written to a file named TCP22 in the C:\PCBIC\LOGS\ subdirectory.

Connect Information

To get connection information about a site, enable this logging option. Items that are logged include:

- ▶ Which Domain Name Server (DNS) is used to resolve an address
- ▶ Whether the connection was successful or not. If it fails, the reason is also written to the log.
- ▶ What port was used for the connection

The following is a sample snippet from an actual log:

```
Looking for bbs.pcboard.com
Using DNS: [206.173.168.40]
Using DNS: [206.173.168.40]
```

```
Looking for bbs.pcboard.com
Connection established on port #23
```

Date & Time

Sometimes the exact date and time that a connection was established proves useful. When you want to know such information, this switch must be enabled. The following is a sample from an actual log:

```
Looking for bbs.pcboard.com
Using DNS: [206.173.168.40]
Using DNS: [206.173.168.40]
Looking for bbs.pcboard.com
Connection established on port #23 (02/27/96 17:34:14)
```

FTP Commands

The FTP application uses several different commands to do things. For example, when a user changes directory the CWD (change work directory) command is sent to the FTP server. This option records all of the FTP commands sent to the server during the connection. The following log snippet illustrates:

```
Sending USER command
Sending PASS command
Cmd: LIST
Cmd: CWD pub
```

FTP Responses

In response to most commands, the FTP server or site sends responses. These responses are preceded with a three digit number corresponding with the type of message it is. With this option enabled, PCB/IC records the first line of each of these messages as shown in this following example:

```
220 www Windows NT FTP Server (Version 3.5).
331 Anonymous access allowed, send identity (e-mail name) as
230 Anonymous user logged in as publicftp (guest access).
200 PORT command successful.
150 Opening ASCII mode data connection for /bin/l.
```

FTP Transfers

A two-stage process is used to get the file from an FTP site to the user. The first stage involves bringing the file from the FTP site to the node machine. From there it can be transferred to the user. PCBoard records the second transfer in the

log because it is a normal file transfer. If you want to record the transfer statistics for the first stage, you must enable this option. An example log entry is as follows:

```
File received: PCBDEMO.ZIP    CPS: 215525
```

Timeout Options

Timeout values determine at what point the Internet application gives up what it is trying to do. Activities such as establishing a connection, receiving input, and inactivity are all covered in this area.

Under the DOS version of PCB/IC, you know when an application is waiting because a visible cue is given. This cue is simply the number of seconds (surrounded by brackets) before the time-out occurs. This example from the FTP application provides a great example:

```
Negotiating login process. [295]
```

From this prompt, we can tell that some sort of negotiation is in process. Furthermore, we also see that if that negotiation does not complete within 295 seconds, the connection will abort. To increase the time-out value temporarily, press the TAB key.

Open Connection

This value determines the maximum time that PCB/IC should wait for a connection response from the host computer. The normal operating procedure is to resolve the address into an IP address, send a connection response and wait for some sort of connection message.

Normally, this procedure happens quite quickly. A slow response time in this area may signal that sending and receiving data may also be slow and therefore not worth keeping the connection. Therefore, a value of 30 is a good default value. If your situation demands more time, by all means, increase it.

Receive Input

When one of the applications is expecting data from the host but is not receiving it, this time-out value is enabled. The application waits until the countdown reaches zero at which time it closes the connection.

There are two main reasons for a delay in receiving characters: 1) there is a delay on the Internet which is usually caused by excessive load, or 2) the host computer is busy processing your request.

In both cases, it is not uncommon to see a delay occur for several moments. A good value for this field is 180 seconds but may vary based on your circumstances.

Transfer Inactivity



This timer simply monitors if there is any data being sent or received. If nothing is sent or received in the time allotted, the connection is automatically closed. Typically, you'll want this to be one of your higher settings (several minutes). Remember, all time-out values are specified in seconds. The following example sets a 5 minute (300 second) value:

```
Transfer Inactivity : 300
```

Close Connection



Just as there is a time-out value for opening a connection, there is one for closing the connection too. In this case, a request to close the connection is made and confirmation is sent in return. If the confirmation is not received in the amount of time specified by this field, the connection is forced closed.

Since closing a connection does not take long, a relatively small value should be entered in this field. A value between 30 and 45 seconds is a good example.

Inactivity Delay



Both the inactivity and receive input delays are quite similar. The major difference is that the inactivity delay is activated whenever the application is not expecting input and is therefore simply idle. If no data is transferred during the time specified by this field, the connection is closed. Set this value to something greater than or equal to your receive time-out value.

Domain Lookup Timeout



Anytime a user enters a named address, the application has to send a query to the domain name server to find out what the IP or numerical address is for that site. These queries are typically answered in just a matter of seconds. It makes sense to set the value for this field to something small (15 seconds or less).

5

Customizing

Life would be boring if everything looked the same. That's one of the reasons we had put a lot of work into allowing you to customize the appearance of PCB/IC. Not only can you change text wording, but you can customize the menu appearance, and many other aspects as well. The focus of this chapter is to detail the areas that can be customized.

Changing Text Displays

Because there are so many ways to convey information, it is important to design systems that are easily customized. So it is with the Internet Collection. Not only can you customize nearly any text prompt that appears, but also add additional text files that enhance the information presented to your callers.

Creating Application Intro/Exit Screens

Although many people are familiar with the World Wide Web, many may not know what Telnet or Ping is. You are given the opportunity to explain these applications in the intro screen for each program. Likewise, additional action may be required at the end of the application where an exit screen can explain this in better detail.

All of these screens we refer to are located in the data file subdirectory (*PCBICCFG | System Configuration*) and can be customized as any other PCBoard display file can. Yes, you can create security specific, or even graphic specific files if desired. About the only thing you cannot do is run a PPE or launch a MNU inside one of these display files.

Intro Screens

Screens that introduce an application serve to prepare the caller for what they are about to enter. For example, if a user selects FTP but does not know what it is, you can have the introduction screen give appropriate background information on the purpose of FTP. Likewise, you may want to include instructions on how to exit the program in case they find out this is not the application they wanted.

The following table outlines information about what intro filenames are associated with which applications. Remember, the location of each display file is in the data file subdirectory as pointed to by PCBICCFG.

Application	Filename	Application	Filename
Finger	FING	RLogin	RLOG
FTP	FTP	SLIP	SLIP
Gopher	GOPH	Telnet	TELN
Ping	PING	TraceRoute	TRACE
PPP	PPP	Whois	WHOIS

As an example, if the data directory is C:\PCBIC\DATA and you want to make an intro screen for FTP, edit the file C:\PCBIC\DATA\FTP.

Closing Screens

Some applications may require additional action by the user. FTP requires that the user download the file from the bulletin board once it is obtained from the FTP server. An excellent way to let the user know what to expect next is through a display file that appears after the Internet application is exited.

At the current time, only the FTP application has a closing screen which consequently is named FTPC. Like the opening screens, this file must be located in the data file subdirectory pointed to by PCBICCFG.

FTP Transfer Window

When a user has ANSI capabilities and when the FTPSCRN file exists, they are shown a progress window similar to the following:

Receiving:		FTP Transfer	File:	of
Time Remaining:	Status:	Average CPS :		
0...10...20...30...40...50...60...70...80...90...100%	Bytes:			
	Done:			

The layout of the sample FTPSCRN file included with the PCB/IC package

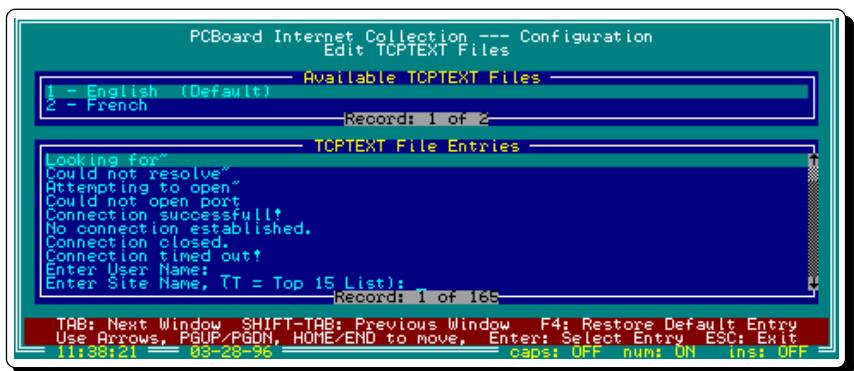
Because this is an external file, you can change the colors of the window using PCBEedit or any of the text if desired. You can even make this window language specific if desired.

The placement of the various options can also be configured by modifying some prompts in the TCPTEXT file. In particular, records 125 to 134 are the ones to be concerned with. The two numbers at the beginning of each line represent the x and y coordinates where each element is placed. For example, to change the location of the filename from column 13, row 2, simply change the numbers for

this record. If you make this change, be sure to update the FTPSCRN file so there is space for the filename to fit.

Individual Text Lines

Virtually everything that is displayed by the Internet Collection is either contained in an external text file or as editable text records in a text record file. To get an idea of the text displayed by PCB/IC, load PCBICCFG and select *Edit Text Records* from the Main Menu.



The initial screen you see when you edit text records.

Notice that this screen is divided into two separate windows. The first shows the language you are editing, and the bottom window provides a scrollable window showing all of the text records you can edit.

Selecting A Language To Edit

When you first enter the text record screen, your current window is the language window. Use the cursor keys to highlight the language you want to edit and press TAB to edit records for that language.

Finding The Text to Modify

Use the cursor keys to scroll through the list of prompts. When you find one to edit, press ENTER to modify it. Let's go through a quick exercise to make sure you understand how to make changes.

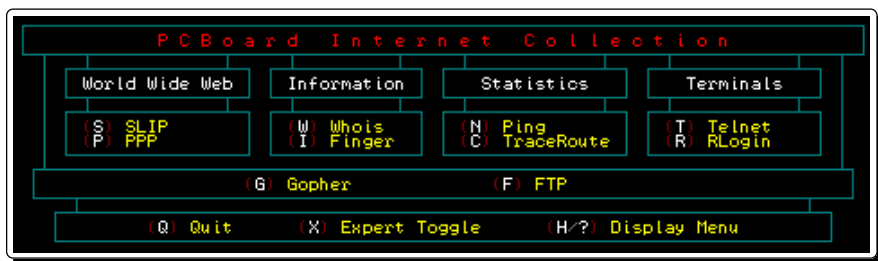
1. Select **Edit Text Records** from the PCBICCFG main menu.
2. Use the cursor keys to select the language to edit.
3. Press TAB to move to the text record window.
4. Use the cursor keys to go down to record 17 and press ENTER.
5. Type in the next text for the record. Press ESC to exit and save.

Saving Changes

PCBICCFG automatically saves the changes when editing text records. Therefore, you do not need to worry about saving your changes.

Menu Customization

The Internet Collection combines the advantages of standard Internet applications into a single easy to use interface. At the heart of this interface is the main menu. This is the list of applications that is shown when you first open the door.



The Internet Collection Main Menu which lists all applications available.

Users can then select what applications they want to access by selecting a single letter or for the more advanced, they can enter the application name followed by the site they want to visit (e.g., `telnet bbs.pcboard.com` allows the user to telnet to Clark Development's BBS.)

When Does The Menu Display?

Like PCBoard, the menu only displays for novice callers and for those who explicitly request it. Novice callers see it every time, experts only see it when they issue the MENU command from the Enter Selection prompt.

Editing The Default Menu

By default, the PCB/IC menu is stored in an external display file just like many of the screens in your PCBoard system. The best utility to use to edit the file is PCBEedit.

The external menu file is stored in the data subdirectory (see *PCBICCFG | System Configuration*) and utilizes the name MENU. Because the MENU file is a simple display file, you can take advantage of PCBoard's display file conventions. Language specific files can be created by adding the appropriate extension to the

end of the file (e.g., MENU.SPA for Spanish callers). Additionally, security specific versions of the file can be created by appending the security to the end of MENU (e.g., MENU40) for security level 40 users.

Modifying The Optional Menu

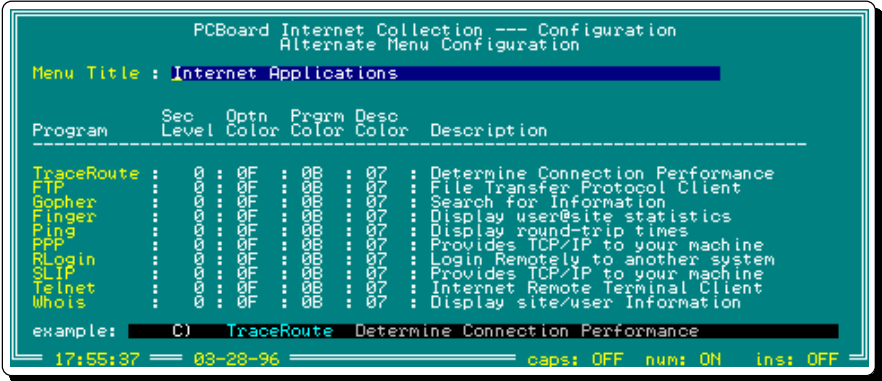
Optionally, you can have PCB/IC automatically create a main menu based on what options they have enough menu security for. The description, menu options, and program name are all automatically colorized and displayed on-the-fly. In order to use this feature, you must delete or rename the external menu file (MENU) located in the \PCBIC\DATA subdirectory.

Changing Application Descriptions

If you want to change the descriptions of the various applications, you can do so easily by selecting **Alternate Menu Configuration** from the PCBICCFG main menu. Use the TAB key to move between fields. When you are at the description field you want to change, simply enter the next text. When you exit the program, be sure to save all of the changes or your menu descriptions will be lost.

Changing Menu Colors

What if you want to change the colors of the menu? Like the descriptions, this too is done by selecting **Alternate Menu Configuration** from the PCBICCFG main menu.



```
PCBoard Internet Collection --- Configuration
Alternate Menu Configuration

Menu Title : Internet Applications

Program      Sec  Optn  Prgrm  Desc  Description
Level       Color  Color  Color
-----
TraceRoute  : 0 : 0F : 0B : 07 : Determine Connection Performance
FTP          : 0 : 0F : 0B : 07 : File Transfer Protocol Client
Gopher      : 0 : 0F : 0B : 07 : Search for Information
Finger      : 0 : 0F : 0B : 07 : Display user@site statistics
Ping        : 0 : 0F : 0B : 07 : Display round-trip times
PPP         : 0 : 0F : 0B : 07 : Provides TCP/IP to your machine
RLogin      : 0 : 0F : 0B : 07 : Login Remotely to another system
SLIP        : 0 : 0F : 0B : 07 : Provides TCP/IP to your machine
Telnet      : 0 : 0F : 0B : 07 : Internet Remote Terminal Client
Whois       : 0 : 0F : 0B : 07 : Display site/user Information

example: C) TraceRoute Determine Connection Performance

17:55:37 03-28-96 caps: OFF num: ON ins: OFF
```

The menu configuration screen in PCBICCFG. This is where menu descriptions and colors can be changed.

Selecting The Item To Change

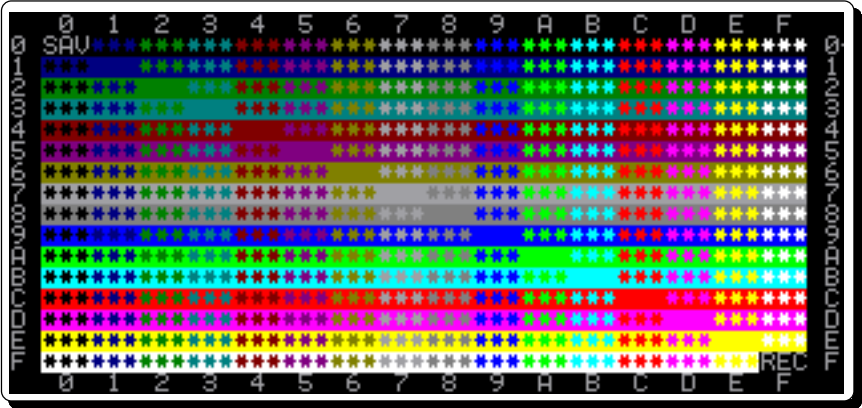
Use the TAB key to select the item you want to change the color of. You can change the color of the menu letter or option, application name, and application description. The following section discusses how to convert the color you want into the appropriate number.

Table Of Colors

The color code for the menu items are the same as the PCBoard color codes covered in your manual only they do not have the @X in front of them. That is, the first digit represents the background color, and the second digit, represents the text color. The following tables illustrate the colors you can select and their corresponding codes:

Color	Background (1st Digit)		Foreground (2nd Digit)	
	Normal	Blinking	Normal	Bright
Black	0	8	0	8
Blue	1	9	1	9
Green	2	A	2	A
Cyan	3	B	3	B
Red	4	C	4	C
Magenta	5	D	5	D
Yellow	6	E	6	E
White	7	F	7	F

A chart showing colors by name and the corresponding letter/digit.



This chart shows the colors you will get. The numbers down the side represent the background colors while the numbers on top and bottom represent the text color. Entries with bright background colors are blinking backgrounds.

6

HotPicks

What Is Their Purpose?

Everywhere you turn it seems that someone is pointing out the top 100 or top sites of the Internet. The purpose of Hotpicks is to allow you to list the top 15 sites for a particular application on your system. That way if you have new callers to the system, they can see a list of frequently visited sites so they can get down to business right away.

Enabling Hotpicks

To allow users to access the hot-pick lists, you must modify their profile. Using PCBICCFG, select *Profile Information*. Notice on the right hand side of the user profile are the following fields:

```
Allow Holding : N
Allow Attach's: N
Hotpicks      : N
```

Use ALT-A to edit one of the profiles. If you want all users to have access to hotpicks, the easiest choice is to modify the DEFAULT group since everyone is a member. Once you modify the group profile to have access to hotpicks, every users that is a member inherits that access.

How Are Sites Added/Tracked?

If the purpose of hotpicks is to track the most common sites, you may be asking yourself "How do sites get tracked?". Any site that has an entry in the hotpicks database is automatically tracked. As a user accesses the site, the number of times the site has been visited is increased.

Whenever a user who has hotpicks access enters a site not in the database, they are asked if they want to add the list to the database. If they answer affirmatively, a description for the site is entered and the database is updated.

Just because a site is added to the database does not mean it will appear for all of your callers. Before being visible the site must be approved. This brings us to our next topic—maintaining the hotpicks list.

Maintaining The Hotpicks List

In maintaining the hotpicks list the primary focus is approving sites for public access and modifying descriptions entered by users. To look at the lists of popular sites for each applications, load PCBICCFG and select *Hotpicks Maintenance*.



The screenshot shows a terminal window titled "PCBoard Internet Collection --- Configuration". Below the title is a section labeled "FTP HotPicks". It contains a table with three columns: "Name", "Description", and "Popularity". The first row is highlighted with a blue background and shows "www.pboard.com" as the name, "PCBoard FTP Site" as the description, and "1" as the popularity. The second row shows "warchive.cdrom.com" as the name, "Walnut Creek's CDROM FTP Site" as the description, and "3" as the popularity. At the bottom of the table, it says "Record: 1 of 2".

Name	Description	Popularity
www.pboard.com	PCBoard FTP Site	1
warchive.cdrom.com	Walnut Creek's CDROM FTP Site	3

Record: 1 of 2

A sample screen showing the hotpicks maintenance for FTP sites.

Notice that this example screen illustrates maintenance for just FTP sites. As we have already discussed, there are lists stored for other applications too.

Switching Between Application Hotpick Lists

To switch between the various application hotpick databases, all you need to do is press TAB or SHIFT-TAB. As you do, notice that the second line changes to show you the application you are currently modifying. It is really that simple.

Approving Sites

The majority of time spent when maintaining a hotpicks database is in approving sites. This action entails getting more information about a particular site, and either approving it or deleting it.

Getting More Information

Use the up and down cursor keys to move between the sites in the database. Once you have a site that has not been approved (is missing the check-mark next to the site name), press ENTER to get more information about it.



The screenshot shows a terminal window titled "Extra Information". It contains a table with two columns: "Field" and "Value". The fields are "IP Address", "Name", "Description", and "Last Used". The values are "206.173.168.40", "www.pboard.com", "PCBoard FTP Site", and "02-28-96" respectively. There is also a "Port: 21" and "Popularity: 1" listed on the right side of the table.

Field	Value
IP Address	206.173.168.40
Name	www.pboard.com
Description	PCBoard FTP Site
Last Used	02-28-96
Port	21
Popularity	1

Information about the PCBoard FTP Site is listed by pressing ENTER.

Making A Site Available

If after looking at the site information, you want to make it available, simply press ALT-A. To let you know that site is now approved a check-mark appears next to the site name. Once the site is approved, it is listed on the hotpicks list for your users.

Deleting Unwanted Sites

When a site is deemed inappropriate, you'll most likely want to delete it from your list. This can be done by pressing ALT-D when highlighting the name. When you delete the site, it can be added again should a user who has hotpicks access go to that site again. To prevent that event from occurring, leave the name in the database but never approve the site.

Automatically Approving All Sites

If you do not want to screen sites before approving them, use the /FORCE command line parameter of the PCBICEVT (event management) program. The proper syntax is:

```
PCBICEVT /FORCE
```

Modifying Descriptions

Not all descriptions entered by the users are appropriate for release. For example, a description may read "Cool site!" That does not offer much to the new user who is looking for a good site to go to.

To edit a description for a site, move the highlight bar using the cursor keys until the desired site is selected and press ALT-E. Enter the new description followed by ENTER to save the changes.

Determining The Most Popular Sites

A tool that you have at your disposal to help you find sites that need to be released is the sort function. Pressing ALT-S while viewing a hotpicks database sorts the database first by unreleased sites and then by popularity of those sites. This way, sites that are the most likely candidates for release are at the top of the list.

The list of the top sites is produced by the PCBICEVT program during your maintenance event. For more information on this aspect of PCBICEVT program, refer to the Maintenance Event section of chapter 2.

The Number of Records in the Database

Each application (Telnet, FTP, etc.) keeps track of a separate database. Each of these databases is capable of tracking up to 2500 sites. To maintain the number of sites you want to track, use the PCBICEVT program as discussed in chapter 2.

7

Program Summaries

Finger

What It Is

This application provides a way for you to find information about a user at a particular site. For example, you can find out the last time a user logged into the system. Many sites have found clever uses for finger requests by having them return additional information such as famous birthdays and the like.

Not all sites accept finger requests. An easy way to determine if a site allows finger access is to try it. If you get a `Host refused connection` message, the site does not support the finger application.

A Sample Finger Site

Let's try a sample finger so you have a feel of what to expect. Begin by selecting Finger from the PCBIC menu or by loading PCBIC with the `/FINGER` command line parameter. The first thing you see is:

```
Enter Name or Site:
```

At the site name, type `cris.com` and press ENTER. For the username, press ENTER again to get a list of users currently on the system.. After the site is queried, you should see output similar to this:

```
[CRIS.COM] - Concentric Research Corporation Network
Home of:
Concentric Research Information Service, BBS Direct and Internet Access.
For subscription information, send an email message to ``INFO@CRIS.COM''
```

To get information about a specific user at that site, enter their name at the "Enter User Name:" prompt. Otherwise, leave it blank to get a list of current users at the site.

When It Does Not Seem To Work

If a site does not have a finger server, you will see a display similar to the following:

```
Enter Name or Site: pcboard.com
Enter User Name:
Looking for pcboard.com
Attempting to open bbs.pcboard.com [206.173.168.20]
```

Notice that the only way you can tell there is no finger information available is that nothing is returned.

FTP

One of the most widely used programs to transfer files over the Internet is called FTP for **F**ile **T**ransfer **P**rotocol. Because these programs are so standard in their interface, we will not go into a discussion of how to FTP a file. Instead, this section focuses on ways that PCB/IC's implementation is different from the others.

Interface

Screen Management

Most FTP programs do not pause when the screen becomes full. FTP uses the traditional PCBoard **More?** prompt to help you see what is on the screen before it scrolls off. When you see this prompt, press ENTER alone to go to the next screen or N followed by ENTER to abort the current display.

Anonymous Logons

Once connected to an FTP server, pressing ENTER at the username prompt attempts to log you into the system as a user named *anonymous*. When you sign in anonymously, sites require your e-mail address for the password. PCB/IC automatically sends this password by taking the user name and adding the e-mail address configured in *PCBICCFG* | *System Configuration*.

Aborting Actions

Another feature that this application inherits from PCBoard is the ability to break out of a display. Press CTRL-K or CTRL-X to abort the current display (e.g., a directory listing.)

File Transfers

Transferring Binary Files

By default, FTP begins in ASCII mode. If you plan to transfer a binary file such as a .ZIP or .EXE file, make sure you get it in binary mode. Otherwise, the file may be truncated early or There are two ways you can do this:

1. Enter the BIN command at the FTP command prompt. This switches all transfers to binary mode.
2. When getting a file, use BGET instead of GET. This forces binary mode to be used.

Getting Multiple Files

This release of PCB/IC does not support multiple file gets. Therefore, use GET and BGET instead of MGET and MBGET.

Gopher

Once connected to a gopher site, navigating it is really quite easy. This is because the very essence of gopher servers is a tiered-menu approach. Once you select an option it either gives you an object of some sort or continues to another menu.

Selecting menu options under PCB/IC's gopher involves entering the corresponding menu letter when prompted. Here is an example:

Cmd	Type	Description
---	-----	-----
A)	Menu	Information About Gopher
B)	Menu	Computer Information
C)	Search	Search Gopher Titles at the University of Minnesota
D)	Search	Search lots of places at the University of Minnesota
E)	Menu	University of Minnesota Campus Information

(1) Quit Enter Selection? a

Two commands are available on nearly every menu. Enter the number 2 to go back to the previous menu (unless you are at the top menu) or enter 1 to quit Gopher.

Ping

What It Is

When you need a quick and easy way to see if you are communicating properly with the Net or if a site is operating, use Ping. With Ping, you are simply given statistics as to the amount of time it took to receive a packet back from a site.

Applying a telephone analogy, this application is the equivalent of calling a phone number, acknowledging that it rung and then hanging up. When you are not connected to the Internet, there is no dial tone so to speak and therefore you cannot verify if a site is operational.

Pinging A Site

Let's ping a popular site like *internic.net* to get a good idea of the output the ping program produces:

```
Enter Site Name: internic.net
Enter number of pings to send (1-9): 3

Pinging internic.net (198.41.0.5) 3 times

sent PING # 1 , response time 162 milliseconds
sent PING # 2 , response time 133 milliseconds
sent PING # 3 , response time 132 milliseconds

Ping Statistics
Sent      : 3
Received  : 3
Success   : 100 %
Average RTT : 142 milliseconds
```

Notable in this output is the response time for each ping as it provides a good performance gauge of your Internet connection. To determine an average response time, simply look at the **Average RTT** field on the last line.

RLogin / Telnet

You should be familiar with this program because it takes a great liking to the popular communications programs like ProComm Plus and others. It is a simple terminal application used to access other computers.

In fact, if you have ever telnetted to `bbs.pcboard.com`, you have contacted our BBS over the Internet instead of phone lines.

The differences between RLogin and Telnet are not that major. The paramount difference is that Telnet can be 7-bit based while Rlogin is guaranteed to be an 8-bit connection.

The default port is 23 for Telnet. If you need to connect using a different port it must be specified after the site address. The proper syntax is `<site_address> <port>`. Notice that there is a space in there. To connect to `bbs.pcboard.com` using port 24, enter `bbs.pcboard.com 24` as the address.

The SysOp can press ALT-F10 from the local keyboard to close the current connection. Any user may abort the current connection by pressing CTRL-].

PPP / SLIP

From the System Operator's perspective, this is probably the plainest application because all it does is forward packets. This simple task, however is what allows users to surf the web because through these applications they establish a temporary connection to the Internet.

Generally, PPP is regarded as the more robust of the two. As such, many users will want to use it.

The SysOp can press ALT-F10 from the local keyboard to close the current connection.

TraceRoute

A tool that will help you determine connection speeds to other sites is the TraceRoute tool. It tells you how many sites (hops) you have to go through to get to another address and also a sample response time in milliseconds.

Five packets are sent to each site in the connection chain. As soon as five packets are returned, the low, high, and average speed of the connection is returned.

```
Enter Name or Site: anysite.com
```

```
Tracing from 206.173.168.20 to another.site.com (168.52.12.26)
Press CTRL-X, CTRL-C or CTRL-K to abort.
```

#)	Router	low high		avg (*)	name
-----	-----	-----	-----	-----	-----
1)	206.173.168.1	2	3	2	206.173.168.1
2)	199.3.96.77	62	66	63	199.3.96.77
3)	199.3.98.42	141	324	211	199.3.98.42

Lower numbers reflect a faster connection between you and the address or site listed. If a number appears in the asterisked column, that means that many packets were lost and had to be resent in order to get five good responses.

When you get asterisks (*) for each of the three values as shown in this example

#)	Router	low high		avg (*)	name
-----	-----	-----	-----	-----	-----
1)	206.173.168.1	*	*	*	206.173.168.1

the site in the **Router** column has not returned 9 successive packets. This may occur because the site is too busy to respond to a lower priority request (TraceRoute request) or the site may be down. Either way, the site is not responding.

Whols

When you want information about a particular domain, you need to use the WHOIS applications to get that information. At the "Enter Name or Site:" prompt, enter the domain to get information about. The following illustrates getting information about **pcboard.com**:

```
Clark Development Co., Inc. (PCBOARD-DOM)
3950 so. 700 E. Suite 303
Murray, UT 84107

Domain Name: PCBOARD.COM

Administrative Contact:
Klingler, Steve (SK79) Steve.Klingler@SALTAIR.COM
(801) 261-1686
Technical Contact, Zone Contact:
Abi-Karam, Ziad (ZA7) ziad@CRIS.COM
408-342-2800

Record last updated on 02-Dec-95.
Record created on 05-Aug-94.

Domain servers in listed order:

NAMESERVER.CONCENTRIC.NET 199.3.12.2
NAMESERVER1.CONCENTRIC.NET 199.3.12.3
```

NOTE: You can also get information about a particular IP address and who is responsible for it by entering the IP address for the site name.

A

When Something Goes Wrong

Errors When Loading

If either the Internet Collection or your network is not configured properly, an error message is usually displayed before being returned to PCBoard. The purpose of this section is to help you understand the error and to outline steps to remedy the situation.

NOTE: In cases where no error message is printed, a more serious error may have occurred. For more details on solving this scenario, refer to the Utilizing Log Files section of this chapter.

NO PACKET DRIVER FOUND

The DOS version of PCB/IC is the only one that uses packet drivers. Essentially, this error occurs when the packet driver signature is not found in memory. To fix this error, you need to verify that the packet driver is loading correctly and is bound to the proper protocols.

ERROR initializing packet driver

The packet driver was found, but it could not be initialized. Sometimes, the driver becomes busy and will fail to communicate. In this case, it is usually best to reset the machine and reload the packet driver.

USERS.SYS not found

PCBoard generates the USERS.SYS file so that PCB/IC can get information about the current caller. If that file cannot be found, it is not possible to run the application. This file is stored in the node subdirectory. Make sure that the entry in DOORS.LST has the USERS.SYS column set to Y.

ERROR accessing packet driver

When this error occurs, it is representative that the packet driver was either not initialized or closed properly. It is not often that this should happen so reboot the affected machine to reload the driver.

Error initializing sockets

The TCP/IP stack could not be initialized. Under DOS, this typically means that some sort of problem occurred with the packet driver or there was not enough memory to load the driver. Under OS/2, the TCP/IP network could not be found or is not completely configured.

ERROR: Opening TCPTTEXT for PCB/IC

The TCPTTEXT file is the file that stores all of the text records that you can customize via PCBICCFG. Obviously, if the file cannot be accessed, the

program will not function correctly. Using PCBICCFG, make sure that the location of these files (Location of Data Files) is a valid location and that the file exists.

ERROR: Reading in USER's TPA

Much of the information about a user (what groups they belong too, how many minutes used in an application, etc.) is stored a special allocation of your PCBoard user file called a TPA. If this area has not been reserved by installing the TPA, PCB/IC cannot load. Use System Manager to make sure the following TPA is setup:

```
Version      :      2
Static Size  :     50
Dynamic Size :      0
Keyword      : PCBIC
```

ERROR: Unable to read in MENU.DAT

When the MENU.DAT file cannot be loaded, PCB/IC safely exits because some security/access information is stored in the file. The location of this file is determined by the Location of Data Files in *PCBICCFG | System Configuration*.

Utilizing The Log Files

One of the resources that is available to you that is often overlooked is the log files. These files often contain status or error messages that are not printed on the screen. Armed with this additional information, troubleshooting can be much, much easier.

A Sample Scenario

For example, let's assume the symptoms are that the program never seems to load. We know the executable is loaded, but we are immediately returned to the bulletin board. What could possibly be the cause. Looking at the log file, we see the following:

```
*****
03-26-96 (16:38) (1) STEVE CATMULL  PCB/IC log file opened
      TPA Error:  Incorrect TPA version
      ERROR:  Initializing TPA routines
      Error writing TPA information
03-26-96 (16:38) (1) STEVE CATMULL  PCB/IC log file closed
```

Notice that there is an obvious problem with the TPA. Looking at the first error message, we see that the incorrect TPA version was found. The signal there is to look into *System Manager | User Info File Maintenance* and make sure the TPA settings are correct.

Finding The Log Files

The log files are in one of two locations. The ultimate location depends on how you answered the **Use External File** question in *PCBICCFG | System Configuration* screen.

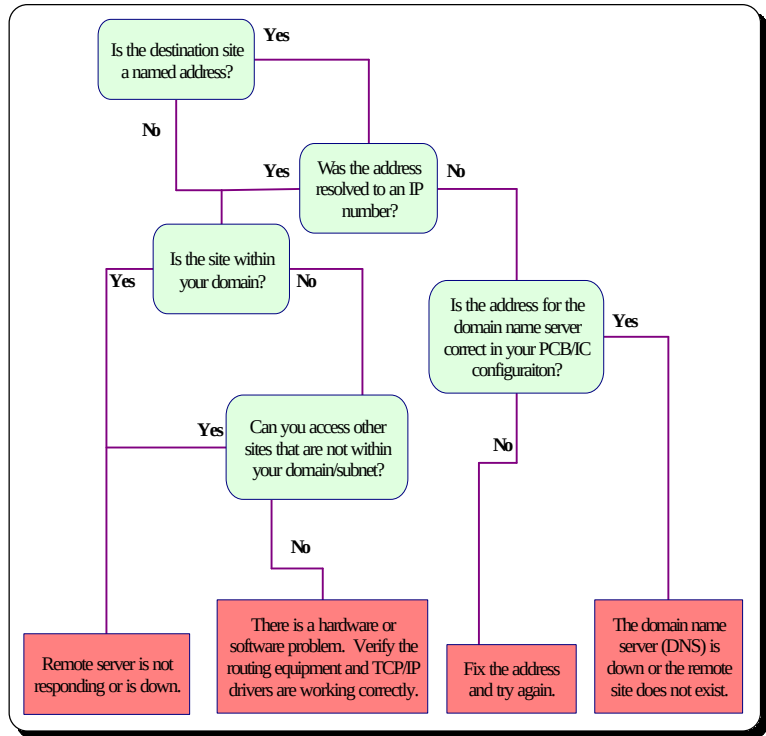
When the external file is in use, log files are created with the name TCP followed by the node number (e.g., TCP1, TCP52, TCP121, etc.) Internal logging stores all of the information directly in the PCBoard caller log.

There is no difference in the content of the information reported in the logs, simply whether the information is put in a separate external log file or directly inside the PCBoard caller logs.

Problems Connecting To Sites

Chances are that if you are in this section, the Internet collection used to work but no longer does or maybe has never worked correctly at all. That is the program runs okay, but you just cannot connect to any sites. The purpose of this section is to help you troubleshoot the situation.

In order to troubleshoot what is going on with the system, you need a basic understanding of what steps take place when trying to contact a site. The following flowchart illustrates how decisions are made when contacting a site. Use it to help you determine where the problem might be.



This flowchart will help you diagnose what has gone wrong.

Let's work through a few sample problems so you get a better feel how to troubleshoot typical scenarios.

Sample Scenario #1

Symptoms: When pinging the site `bbs.pcboard.com`, the message could not resolve `bbs.pcboard.com` appears.

Solution:

1. This is a named site.
2. The site `bbs.pcboard.com` was not resolved to an address so check the DNS or name server address in `PCBICCFG`.
3. If the DNS address is correct, check to make sure your DNS is functioning. It may either be down or inaccessible for whatever reason. A good way to test the inaccessible possibility is to go to a known good IP address to see if you can access other sites.

Sample Scenario #2

Symptoms: When connecting to ftp.pcboard.com the message Host refused connection.

Solution:

1. This is a named site.
2. We know that the site name was resolved to an IP address because we see the [206.173.168.20] on the Attempting to open line.
3. Referring to the chart we make note that this site is not in our domain, so we move on to the next step—checking another site.
4. We attempt to open ftp.cdrom.com and are successful. Therefore, the first site must be temporarily down.

General Technical Support Tips

Most problems that you experience with computer related topics can be better understood and consequently fixed faster if you adhere to three principles. In this section we outline the three founding principles that will aid you in your system

Duplicate The Problem

The ability to duplicate a problem consistently is the first step to finding a solution. Only when you can reliably duplicate the problem will you be able to tell if a modification has taken care of it.

When trying to duplicate an event, try to recreate the environment. Use the same user record and, where possible, use the same keystrokes. Consistency is the name of the game when duplicating a problem.

Simplify The Environment

Once you have duplicated the problem reliably, the next area of focus needs to be in simplifying the system. Remove any un-necessary TSRs or hardware in the machine.

In your efforts to simplify the system, make sure you make only one change at a time. After each change, try to duplicate the problem. If the symptoms do not occur, the problem is most likely taken care of. You'll know by performing follow-up.

Follow-Up

With a hypothesis of what is causing the problem, put the cause of the problem back into the system. What you are hoping to see is the problem occur again. If the problem is not exhibited, the cause of the problem has been improperly identified. Keep simplifying the system and remove what you can. This may involve running one node when you have, or running just from DOS rather than from another operating system.

Problems Running Under DESQview

We like to use our beta test cycles to help locate any potential problems with the diversity of systems that PCBoard and associated utilities can run on. It was during one of these testing cycles that we were made aware of a potential problem under DESQview revolving memory manager exception errors.

If you are experiencing these errors, a change must be made to the performance settings under DESQview. Using DVSETUP, configure the network buffers to be 5 times the number of nodes (e.g., 20K of network buffers for 4 nodes).

Problems Running Under OS/2

Proxy ARP Failure

This error message is only displayed during PPP and SLIP connections. Although it appears on your screen, it has no effect on the connection or ability to communicate. Therefore, simply ignore the message.

PPP And SLIP Do Not Work But Other Applications Do

In this case, chances are that you have set the IP address environment variable incorrectly. Remember that each node needs to be assigned a unique IP number that you really own or are authoritative for. Otherwise, your users will be able to send requests, but never receive them.

An additional item to be aware of is that the IP addresses you use for the nodes should not be equivalent to the IP address assigned to the machine. Make sure they are unique.